# Practical 4: Build a RetailPulse Region × Product Crosstab

Related topic: Create Crosstab Reports

## Objective

Construct a crosstab report summarizing RetailPulse sales across Region and Product Category with multiple measures and summary totals.

## Materials required

IBM Cognos Analytics access and the RetailPulse Sales package.

## Step-by-step procedure

| Step | Action in Cognos Analytics | What you should see |
|---|---|---|
| 1 | Choose **New > Report**, select the RetailPulse Sales package and the **Crosstab** template. | An empty crosstab with a row edge, a column edge and a body is shown. |
| 2 | Drag **Region** to the row edge and **Product Category** to the column edge. | Regions run down the side; the five categories run across the top. |
| 3 | Drag **Sales Amount** into the crosstab body. | Every Region × Category cell is populated with a summed amount. |
| 4 | Drag **Quantity** next to Sales Amount (drop on the right edge of Sales Amount) to nest it as a second measure. | Each category column splits into Sales Amount and Quantity. |
| 5 | Select Region and choose **Summarize > Total**; select Product Category and choose **Summarize > Total**. | A Total row (column summary) and a Total column (row summary) are added. |
| 6 | Select Sales Amount, set **Data format > Currency**; click **Run**; **Save as** “RetailPulse Region-Product Crosstab”. | Sales Amount shows $ values; the report is saved in My content. |

## Sample solution

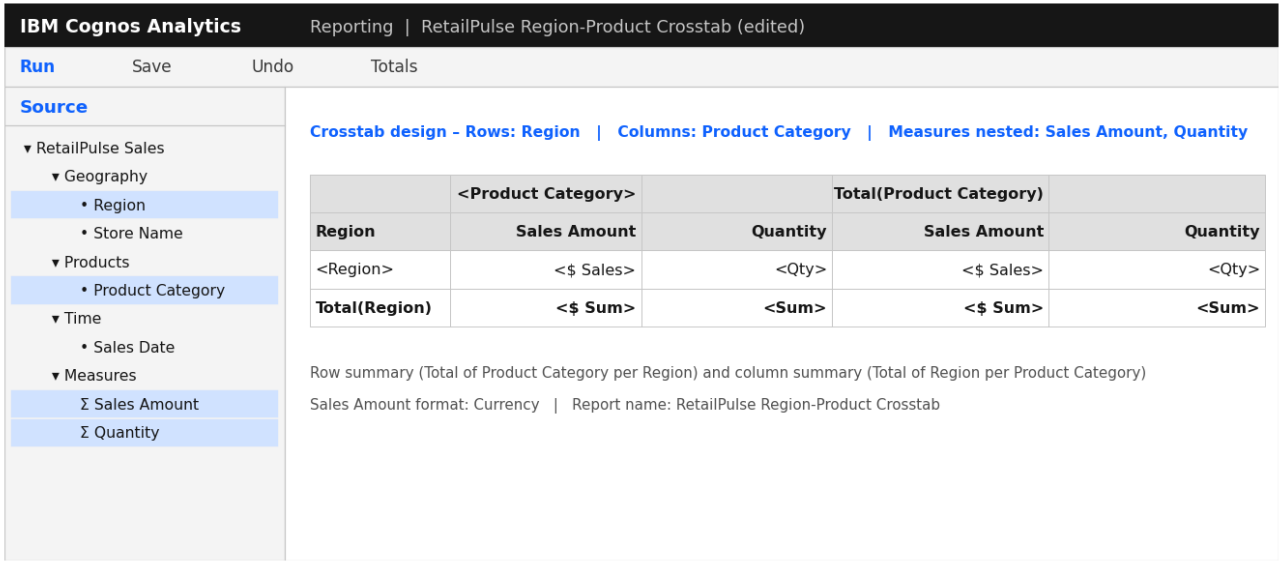



Figure 1. Crosstab design: Region (rows), Product Category (columns), nested measures and summary totals.

## RetailPulse Region-Product Crosstab

| | Apparel | | Electronics | | Grocery | | Home & Kitchen | | Sports & Outdoors | | Total | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Region** | **Sales Amount** | **Quantity** | **Sales Amount** | **Quantity** | **Sales Amount** | **Quantity** | **Sales Amount** | **Quantity** | **Sales Amount** | **Quantity** | **Sales Amount** | **Quantity** |
| **North** | $210,857 | 4,396 | $291,363 | 2,009 | $177,580 | 12,695 | $192,071 | 3,090 | $182,970 | 2,542 | $1,054,842 | 24,732 |
| **South** | $246,974 | 5,149 | $329,103 | 2,272 | $201,155 | 14,351 | $232,257 | 3,742 | $219,361 | 3,043 | $1,228,850 | 28,557 |
| **East** | $181,790 | 3,799 | $233,157 | 1,611 | $151,860 | 10,853 | $165,462 | 2,660 | $165,688 | 2,304 | $897,957 | 21,227 |
| **West** | $266,082 | 5,531 | $358,363 | 2,479 | $219,772 | 15,677 | $253,744 | 4,088 | $241,477 | 3,347 | $1,339,438 | 31,122 |
| **Total** | **$905,704** | **18,875** | **$1,211,986** | **8,371** | **$750,368** | **53,576** | **$843,533** | **13,580** | **$809,497** | **11,236** | **$4,521,087** | **105,638** |

Grand total: $4,521,086.88 sales amount and 105,638 units, summed across 4 regions and 5 product categories.

## Reconciliation with the grouped list report (Practical 2)

| Region | Crosstab row total (Sales Amount) | Grouped list region total (Practical 2) | Match |
|---|---|---|---|
| North | $1,054,842.04 | $1,054,842.04 | Yes |
| South | $1,228,849.52 | $1,228,849.52 | Yes |
| East | $897,957.47 | $897,957.47 | Yes |
| West | $1,339,437.85 | $1,339,437.85 | Yes |
| All regions | $4,521,086.88 | $4,521,086.88 | Yes |

| Expected outcome check | Result |
|---|---|
| Regions and categories align without misalignment | **Pass** – 4 regions x 5 categories, 2 measures each |
| Each cell shows the correct aggregated measures | **Pass** – Cell values equal Practical 2 category subtotals per region |
| Row and column totals sum correctly | **Pass** – Grand total $4,521,086.88 / 105,638 units |
| Sales Amount consistently formatted as currency | **Pass** – All Sales Amount cells show $ |
| Totals reconcile with the earlier grouped list | **Pass** – Practical 2 overall total = $4,521,086.88 |

Note on screens: the screenshots in this document are Cognos-style renderings produced from a synthetic RetailPulse dataset (12 stores, 4 regions, 5 product categories, 1 Jan–25 Oct 2026). Replace them with live captures from your own Cognos environment if your submission requires real screenshots.